

Math100C - WK8

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1.

Let n be a positive integer. Define the polynomials

$$s_i = u_1^i + \dots + u_n^i, \quad (i \in \{1, \dots, n\})$$

Prove that these generate the ring of symmetric polynomials in $u_1, \dots, u_n$ over $\mathbb{Q}$ but not over $\mathbb{Z}$ when $n \geq 2$.

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 Newtons' identities (plural?) gives a recursive formula for calculating ke_k , $k \in \{1, \dots, n\}$ from $s_1, \dots, s_k$ and $e_1, \dots, e_{k-1}$:

$$ke_k = \sum_{i=1}^k e_{k-i} s_i \quad (1)$$

Over $\mathbb{Q}$, the coefficient k on the left hand side of (1) can be canceled, so Newton's identities says all the elementary symmetric polynomials $e_1, \dots, e_n$ can be written in terms of $s_1, \dots, s_n$; the elementary symmetric polynomials generate the ring of symmetric polynomials over coefficients in any ring R , so $s_1, \dots, s_n$ generate the ring of symmetric polynomials over $\mathbb{Q}$.

I don't really know how to proceed for the the symmetric polynomials over $\mathbb{Z}$.

2.

Let n be a positive integer and G a subgroup of $\mathfrak{S}_n$ containing at least one transposition.

(a)

Suppose G contains all the transpositions of $\mathfrak{S}_n$. Prove $G = \mathfrak{S}_n$.

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 Since every permutation in $\mathfrak{S}_n$ has a unique cycle decomposition, it suffices to show that G contains all the cycles of $\mathfrak{S}_n$. To do this, induct on cycle length k . $k = 2$ holds by the assumption on G . Suppose G contains all cycles of length less than k . Since relabeling the elements $1, 2, \dots, n$ doesn't affect the inductive hypothesis, showing that G contains all k -cycles is equivalent to showing that G simply contains $(1\ 2 \cdots k)$. Well, by the inductive hypothesis, $(1\ 2 \cdots [k-1])$, $([k-1]\ k) \in G$, so

$$(1\ 2 \cdots [k-1]) \cdot ([k-1]\ k) = (1\ 2 \cdots k) \in G,$$

finishing the proof.

(b)

Suppose $n = p$ is prime and $p \mid \#G$. Prove $G = \mathfrak{S}_n$ by reducing to the previous question.

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I first prove the lemma Kiran suggested:

Lemma. *Let a subgroup $G < \mathfrak{S}_n$ correspond to a graph $\Gamma = (V, E)$ with vertices $1, \dots, n$ and edge ij whenever G contains a transposition (ij) . If Γ is connected, then $G = \mathfrak{S}_n$.*

Proof of Lemma. First, consider the special case where the vertex 1 is connected to all other vertices, i.e., G contains all transpositions $(1k)$ for $k \in \{2, \dots, n\}$. Then G contains all the transpositions $\mathfrak{S}_n$: indeed, for any $i \neq j \in \{1, \dots, n\}$,

$$(1i)(1j)(1i) = (ij)$$

By part (a), $G = \mathfrak{S}_n$.

The general case will reduce to this special case. Given any $k \in \{2, \dots, n\}$, $(1k) \in G$ by the following: since Γ is connected, there is a walk

$$1 \rightsquigarrow u_1 \rightsquigarrow \dots \rightsquigarrow u_t = k$$

But the calculation

$$(1u_1)(u_1u_2)(1u_1) = (1u_2)$$

shows there is a direct path $1 \rightsquigarrow u_2$ in Γ . Inductively performing this calculation, one simply obtains a direct path $1 \rightsquigarrow k$, i.e., $(1k) \in G$, as was to be shown. $\square$

Let G be a subgroup of $\mathfrak{S}_p$ such that it contains a transposition and $p \mid \#G$. By Cauchy's theorem, G contains an element of $\mathfrak{S}_p$ of order p . The only such element is a p -cycle; by relabeling the p -cycle as needed, one can assume that G contains $(12 \cdots p)$ and some transposition (ij) .

$$(12 \cdots p)(ij)(1p[p-1] \cdots 2) = ([i+1][j+1])$$

shows G contains all transpositions of the form $([i+t][j+t])$. Namely, it contains $(1k)$ for some $k \in \{2, \dots, p\}$ and therefore $([1+t][k+t])$ for all t . If Γ is the graph corresponding to G as in the lemma, the notice that Γ has vertices

$$1 \rightsquigarrow k \rightsquigarrow (2k-1) \rightsquigarrow (3k-2) \rightsquigarrow \dots$$

Hence if

$$x \mapsto xk - (x-1) = (k-1)x + 1$$

is surjective mod p , then Γ is connected. Indeed, the map is surjective mod p because $k-1 \not\equiv 0 \pmod{p}$ and linear equations always have a solution mod p (since $\mathbb{F}_p$ is a field). Hence Γ is connected, and the lemma then gives $\mathfrak{S}_p = G$.

3.

Let f be a polynomial of degree n over a field F . Prove that a splitting field for f over F has degree dividing $n!$

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Let K/F be the splitting field for the polynomial f over F of degree n . Letting $G := \text{Gal}(K/F)$, recall K/F is a Galois extension, $K^G = F$, and $[K : K^G] = \#G$. Since K/F is Galois, G acts on the roots of f faithfully, giving

$$G \hookrightarrow \mathfrak{S}_n$$

hence $\#G \mid \#\mathfrak{S}_n = n!$. Putting together all the facts,

$$[K : F] \mid n!,$$

as was to be shown.

4.

Compute the Galois group of the splitting field $x^5 - 2$ over $\mathbb{Q}$.

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The splitting field for $f(x) := x^5 - 2$ over $\mathbb{Q}$ is

$$\mathbb{Q}(\zeta_5, 2^{1/5}) :$$

ζ_5 and $2^{1/5}$ clearly generate all the roots of f ; meanwhile, any field containing all the roots of f clearly contains $2^{1/5}$ and $\zeta_5 2^{1/5}$, and multiplying the inverse of the former with the latter gives ζ_5 in that field as well.

Let $K := \mathbb{Q}(\zeta_5, 2^{1/5})$. The minimal polynomial for $2^{1/5}$ over $\mathbb{Q}$ is f , a degree 5 polynomial, whereas ζ_5 is a degree 4 element over $\mathbb{Q}$. Since these are coprime,

$$[K : \mathbb{Q}] = 20.$$

Finally, it is clear f is irreducible, for example by Eisenstein for $p = 2$, so the Galois group $G := \text{Gal}(K/\mathbb{Q})$ is the unique transitive subgroup of $\mathfrak{S}_5$ of order 20.

Let's compute this group. Associate the roots of f $2^{1/5}, 2^{1/5}\zeta_5, \dots, 2^{1/5}\zeta_5^4$ with indices $1, 2, \dots, 5$, respectively. First, consider the permutation $\sigma \in G$ mapping

$$\zeta_5 \mapsto \zeta_5^2, \quad 2^{1/5} \mapsto 2^{1/5}.$$

One can easily check that σ has order 4 in G and acts on the roots of f as **(2354)**. Next, consider $\tau \in G$ mapping

$$\zeta_5 \mapsto \zeta_5, \quad 2^{1/5} \mapsto 2^{1/5}\zeta_5.$$

It is clear that τ is order 5 in G acting on the roots of f as **(12345)**. A 5-cycle and a 4-cycle generates a group of order 20, i.e.,

$$G = \langle \sigma, \tau \rangle.$$

This computes the Galois group for the splitting field of $x^5 - 2$ over $\mathbb{Q}$.

5.

Prove that $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$ is a Galois extension of $\mathbb{Q}$ and compute its Galois group.

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 Let $K := \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$. The generators of K over $\mathbb{Q}$ are precisely the roots (up to $\pm$) of

$$f(x) := (x^2 - 2)(x^2 - 3)(x^2 - 5),$$

so K is a splitting field over $\mathbb{Q}$, i.e., $K/\mathbb{Q}$ is a Galois extension. We've shown many times in class that $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ is a degree 4 extension of $\mathbb{Q}$, and it's clear $\sqrt{5} \notin \mathbb{Q}(\sqrt{2}, \sqrt{3})$, so

$$[K : \mathbb{Q}] = 8.$$

Let $G := \text{Gal}(K/\mathbb{Q})$. By the paragraph above, G is a subgroup of $\mathfrak{S}_6$ of order 8. Since K is the splitting field for f , any $\sigma \in G$ permutes the roots of each factor of f :

$$\sqrt{2} \mapsto \pm\sqrt{2}, \quad \sqrt{3} \mapsto \pm\sqrt{3}, \quad \sqrt{5} \mapsto \pm\sqrt{5}$$

None of the factors of f share a root, so each of these three choices can be made independently; i.e., the permutations of G sending

- $\sqrt{2} \mapsto -\sqrt{2}$, identity on other elements
- $\sqrt{3} \mapsto -\sqrt{3}$, identity on other elements
- $\sqrt{5} \mapsto -\sqrt{5}$, identity on other elements

correspond to disjoint 2-cycles, hence they generate $(\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$, which has order 8. Hence $G \cong (\mathbb{Z}/2\mathbb{Z})^3$.

6.

Define a field $K = \mathbb{C}(t)$. Define automorphisms σ, τ of K over $\mathbb{C}$ mapping

$$\sigma(t) = \frac{t+i}{t-i}, \quad \tau(t) = \frac{it-i}{t+1}$$

Prove these automorphisms generate a group isomorphic to A_4 and compute the fixed field.

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 I utilize the isomorphism $\text{GL}_2(\mathbb{C})/\mathbb{C} \cong \text{Aut}(\mathbb{C}(t)/\mathbb{C})$. Identify

$$\begin{aligned} \sigma(t) &\rightsquigarrow M_\sigma := \begin{bmatrix} 1 & i \\ 1 & -i \end{bmatrix} \\ \tau(t) &\rightsquigarrow M_\tau := \begin{bmatrix} i & -i \\ 1 & 1 \end{bmatrix}. \end{aligned}$$

Then

$$\begin{aligned} (M_\sigma)^2 &= \begin{bmatrix} 1+i & 1+i \\ 1-i & i-1 \end{bmatrix}, & (M_\sigma)^3 &= \begin{bmatrix} 2+2i & 0 \\ 0 & 2+2i \end{bmatrix} = I_2 \\ (M_\tau)^2 &= \begin{bmatrix} -1-i & 1-i \\ i+1 & 1-i \end{bmatrix}, & (M_\tau)^3 &= \begin{bmatrix} 2-2i & 0 \\ 0 & 2-2i \end{bmatrix} = I_2 \end{aligned}$$

So $(M_\sigma)^3 = (M_\tau)^3 = 1$. Also,

$$M_\sigma M_\tau = \begin{bmatrix} 2i & 0 \\ 0 & -2i \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

shows $M_\sigma M_\tau \neq I_2$ and $(M_\sigma M_\tau)^2 = I_2$. This means M_σ and M_τ satisfy the relations for the presentation

$$A_4 = \langle x, y \mid x^3 = y^3 = (xy)^2 = 1 \rangle$$

In other words, σ and τ generate A_4 .

I'm unsure what the fixed field is. I suspect it is some sum of the matrices computed above.

7.

Let $\zeta_n := e^{2\pi/n}$ for some positive integer n . Prove $\mathbb{Q}(\zeta_n)$ is a Galois extension of $\mathbb{Q}$ with Galois group isomorphic to $(\mathbb{Z}/n\mathbb{Z})^\times$.

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Let $G = \text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$. Recall that the minimal polynomial for ζ_n over $\mathbb{Q}$ is

$$\Phi_n(x) = \prod_{\substack{a < n \\ (a,n)=1}} (x - \zeta_n^a) \quad (2)$$

Thus it is clear that $\mathbb{Q}(\zeta_n)/\mathbb{Q}$ is a splitting field for $\Phi_n(x)$ over $\mathbb{Q}$, i.e., a Galois extension of $\mathbb{Q}$: it contains all the roots of $\Phi_n(x)$, while obviously, any field containing all the roots of $\Phi_n(x)$ contains ζ_n . (2) also shows that any $\sigma \in G$ maps

$$\zeta_n \mapsto \zeta_n^a$$

for some $a < n$, $(a, n) = 1$. Hence the following map $G \rightarrow (\mathbb{Z}/n\mathbb{Z})^\times$ is well defined:

$$\sigma \mapsto a \quad (3)$$

where a is such that $\sigma(\zeta_n) = \zeta_n^a$. This map is clearly injective since σ is defined entirely by how it sends ζ_n , which means only one $\sigma \in G$ sends ζ_n to ζ_n^a for a fixed a . The map is also a homomorphism: given $\sigma_1, \sigma_2 \in G$ with $\sigma_1(\zeta_n) = \zeta_n^{a_1}$ and $\sigma_2(\zeta_n) = \zeta_n^{a_2}$,

$$\begin{aligned} \sigma_1 \sigma_2(\zeta_n) &= \sigma_1(\zeta_n^{a_2}) \\ &= (\sigma_1(\zeta_n))^{a_2} \\ &= \zeta_n^{a_1 a_2} \end{aligned}$$

Finally, because $\mathbb{Q}(\zeta_n)/\mathbb{Q}$ is a splitting field, $\#G = \varphi(n)$. $\varphi(n) = (\mathbb{Z}/n\mathbb{Z})^\times$ as well, so (3) is surjective; this finishes the proof that $G \cong (\mathbb{Z}/n\mathbb{Z})^\times$.

8.

Let $f(x) = x^3 + x + 1$. Prove the splitting field of f over $\mathbb{Q}$ contains $\sqrt{-31}$.

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 I read about the discriminant not in Artin, but in Dummit and Foote because the latter makes more sense to me. In general, for any f , the definition of the discriminant gives

$$\sqrt{D} = \prod_{i < j} (\alpha_i - \alpha_j)$$

where the α_i and α_j 's are roots of f , which shows $\sqrt{D}$ is always in the splitting field for f . For $f(x) = x^3 + x + 1$, the discriminant is

$$D := -4(1)^3 - 27(1)^2 = -31.$$

Hence $\sqrt{-31}$ is in the splitting field for f over $\mathbb{Q}$.